

Title:

The Evolution of SMiShing: A Literature Review of Coordinated Threats, Psychological Triggers, and Multi-Domain Defense Strategies

Abstract

(what is the difference between an abstract, introduction, and conclusion? Don't they all state the problem, and the major talking points?)

SMS Phishing aka SMiShing has evolved from simple messages on a mobile phone to multi-domain coordinated attacks. In fact, the Federal Trade Commission has reported losses increasing by over 450% since 2020 to about \$470 million. Research shows that, as with email phishing and other scams, smishing is often targeted and coordinated by professional attackers. However, it remains unclear to what extent the cybersecurity field has focused on combatting smishing as a distinct and widespread cybercrime. A systematic literature review can shed light on different types of studies and draw connections among many fragmented research efforts. In this project, we conduct such a review. First, we established a good baseline by reading and reviewing news articles on the topic of smishing. Then, we used keywords drawn from those articles to search databases such as Google Scholar. We diversified our knowledge base by looking at other countries for smishing news and research. We also added keywords to find recent papers and projects to mitigate smishing, such as those that use machine learning or that include educational solutions. We read the sources, made a literature matrix and notes sheet to break the sources' information into topics and to write down observations and reflections. Based on our findings, we recommend more interdisciplinary research between education and technical specialists. This would produce awareness programs and mobile-first interventions that can address smishing and wider coordinated attacks. This work is a first step towards meaningful action against mobile cybercrime and increasing trust in U.S. telecom.

Introduction

SMS phishing, also known as SMiShing, has become a worldwide cybersecurity threat. This threat has evolved from simple deceptive text messages to threat actors using multi-domain coordinated attacks that involve multiple platforms and infrastructures that affect the cyber ecosystem. Unlike traditional phishing attacks, smishing has become more prevalent overall due to immediate perceived trustworthiness that texting services once offered. This perceived trust in text messages has had financial effects including consumers reporting losing over \$470 million dollars to scam texts alone in 2024 [12].

Research has consistently shown that smishing campaigns are not random and uncoordinated in nature. Smishing has evolved to large illegal opportunistic driven activity that resembles structured criminal enterprises that operate at a large scale with professional habits. Some studies have noted structured bot operations, large-scale brand impersonations, and cross channel coordinated attacks that involve smishing, vishing, and traditional phishing campaigns targeting the same person or business[3, 7]. These campaigns frequently reuse domains, malicious links, and often impersonate well-known brands in order to build a facade of credibility.

At the same time, threat actors utilize and heavily rely on basic human psychology to trigger emotions to get a response. Threat actors create deceptive messages invoking emotions such as urgency, authority bias, and trust through the messages that are sent. These psychological triggers increase the likelihood that recipients will click the malicious links or respond with sensitive information[5].

The following literature review examines a sample and overview of the smishing research that has been conducted over the last decade. It explores how the threat operates, the common themes discovered across existing studies, and evaluates the technical and educational solutions that have been proposed by researchers to reduce the impact the public faces from these attacks.

Motivation and Research Question(s)

Talk a bit more about why this review is needed

Then state the implicit RQ or several RQs or whatever is appropriate.

This literature review is guided by the following research questions (RQs):

1. How are modern smishing campaigns structured and coordinated across the digital infrastructure?
2. What are the psychological and demographic factors that influence a user's susceptibility to smishing attacks?
3. How effective are current machine learning and technical detection systems in mitigating smishing?
4. What gaps exist in current smishing defenses, and what interdisciplinary strategies are needed to address them?

These research questions aim to help bridge the gap between fragmented studies across technical, behavioral, and educational research related to the smishing problem. The motivation for this literature review was guided by the lack of understanding of why smishing still exists while phishing attacks are often filtered out. It seems to be that there is no filtering for text messages which often leaves a gap that attacks often exploit in phishing type attacks.

Methodology

Throughout this literature review, I used a systematic approach so that every paper had the same way of being processed. All academic sources were identified using Google Scholar during the literature search. Zotero was used for collecting, managing, citing, and reading the sources.

The most common keywords that I used were "Smishing" and "Smishing Attack". Sources were chosen for a variety of different reasons. These reasons included popular cited articles, titles

mentioning different types of smishing attacks, and smishing focused in other parts of the world to obtain a global perspective of the smishing issue.

The systematic approach used for analyzing each source included the following steps: locating the source using the keywords on Google Scholar, storing the article in Zotero, reading the article, taking structured notes in Google Docs, then compiling the data in five categories in a Google Sheets literature matrix. The matrix categories are ACM citation, goal of the paper, key findings, future work recommendations, and any additional notes or gaps found from the paper. These categories were decided because they would each provide the necessary information to write this paper and understand the current landscape of smishing based on the sources. Each category would be referenced for each paper mentioned in this review. ACM citation is used for the bibliography, goal of the paper was used to understand what the author is trying to accomplish, key findings were selected so I could compile the most important information (statistics, major findings, etc.), future work was decided as a category so we could see where the researcher recommends and so we can see their future smishing work), and notes and gaps were created to help find the information and research gaps that needed to be identified.

[How did you decide on the categories?]

This method of organization was used for further analysis and compilation of different products including a slideshow for presentation and this current paper. This also allowed for patterns and recurring themes to emerge across the multiple studies.

Literature Themes

1. Smishing is coordinated, structured, and systematic in practice

Multiple studies have indicated that smishing campaigns operate at a scale similar to small organized businesses and exhibit high structural coordination within their own practices. Recent literature as of 2025 increasingly suggests that smishing is not a series of isolated scam texts, but a part of a structured cybercrime ecosystem. There are multiple examples of recurring patterns of organization, template and website reuse, impersonations, and operational consistency that resembles an organized campaign geared towards scam texts rather than random opportunistic attacks. This is an important feature of smishing because it challenges how smishing needs to be conceptualized. Instead of treating each message as a standalone attempt of malicious behavior, it should be thought of as a broad, structured, and adaptive criminal system that aims to steal personal information for financial gains.

This broad view is visible in work that examines the relationship between messages instead of singularly classifying each message to see if it has malicious roots or not. SmishViz data has shown that many campaigns reuse similar domains and message structures. A study examining SmishViz data found that 88.6% of smishing messages contained URLs and 65.9% impersonated legitimate brands. 182 unique brands were used in the smishing campaigns from their dataset which included financial institutions, delivery services, large technology companies, and more. The study also found that threats often change their smishing messages by

frequently reusing and creating new domains and utilizing URL shortening services to hide the malicious original website link [7].

Additional research has also identified the structured operational patterns among smishing botnets. Saeki et al. (2022)[6] documented these predictable patterns within smishing bot activity targeting users that are in Japan. The study estimated that 90% of botnet smishing messages were coming from two major types of bots. The analysis concluded that these bots were consistently updated by software developers, operated in state-like patterns (sleep, silent, and active modes). The researchers also observed that software updates to these bots followed patterns similar to normal work schedules. What is also important to note is that over a long period of time, updates are never pushed from 12 pm - 1pm indicating lunch time, updates slowed down during night time hours, and updates were never pushed on Sundays. For updates, they frequently consisted of updating the smishing link itself, which would take potential victims to different websites. These patterns have a strong indication and further support the idea that smishing organizations are not rudimentary, but structured, organized, and efficient in mirroring regular professional business hours. The threat actors and their smishing campaigns may operate at a coordinated criminal enterprise rather than simple opportunistic scams.

This theme also shows an important gap in older and narrower defenses. Many anti-smishing studies focus on message-level detecting, but the studies should switch from the message level to the ecosystem level. If attackers reuse domains, infrastructure, timing patterns, and impersonation themes across many messages, then an effective defense requires analysing threat relationships and the whole cyberecosystem. In this case, the literature would move beyond an idea of simple mobile phishing to building a frame of a scalable ecosystem level threat that requires ecosystem analysis and response.

2. Human psychology is a driving factor in smishing success

A second major theme across the literature is that smishing not only succeeds because of technical deception, but also because it exploits predictable aspects of human judgement and emotion. A user response plays a central role in whether or not a smishing attack succeeds. This shifts smishing from only being a cellular and computer issue to a behavioral problem as well. Several studies are able to reinforce this point by showing that users do in fact respond to smishing when the message is curated in the correct way.

Rahman et al. (2023)[5] conducted an experimental study examining user responses to smishing messages and found that 16.92% of participants responded to smishing during the first round of experiments and 12.82% responded in a second round. A response was identified if the user clicked on the link, replied to the message, or called the phone number in the message. Some users interacted with the message even if they recognized that it was a smishing attempt out of curiosity. Curiosity and uncertainty were also frequently cited as reasons for interacting with suspicious messages.

Curiosity isn't the only thing that may cause persons to interact with smishing messages. Other research highlights how demographic and social factors influence susceptibility to smishing attacks and how threat actors use this to their advantage. In a study conducted by Tabassum et al. (2025)[9], international students (referred to as educational migrants) living within the United States were particularly vulnerable and had trouble with smishing messages due to the authoritative impersonation. The IRS, DHS, housing, and immigration services were known to be impersonated by attackers. These impersonations leveraged authority and fear to pressure recipients to respond quickly to the smishing attempts[8].

Targeted demographics may also influence the rate of susceptibility. Rahman et al. (2023)[5] observed that African American and Mexican American participants typically had a higher response rate to the smishing messages in the United States. However, this could possibly be factored for other countries as well. For example, Zimba et al. (2025)[11] conducted a smishing study in Zambia and found that 93% of his participants have received a smishing message at some point in time and 63% were unaware and unfamiliar of the concept of smishing. Zimba et al. also noted that the smishing messages received from his participants in Zambia found a common theme of fear, urgency, and greed that was used in the messages in an attempt to collect personal information or financial information. These findings highlight the importance of understanding smishing as both a technical and behavioral cybersecurity problem.

An important understanding of this area of smishing literature is that technical defenses alone are unlikely to solve the smishing issue. This problem is severe when users continue making decisions under pressure, ambiguity, or misplaced trust. The literature shows that smishing relies on psychology and is not a secondary issue. It is a central issue and due to way that attackers rely and manipulate authority, urgency, and familiarity. Effectively, future solutions must include and focus on the users and account for human behavior, not just malicious code and suspicious links.

3. New Machine learning systems show promise but cannot keep pace with the evolving threat

A significant portion of recent smishing research focuses on the development of machine learning models designed to detect possible smishing messages to warn users. These systems typically analyze message content, URLs, and other structural features to identify patterns associated with common smishing attacks. This new technical defense has made significant progress in smishing due to their strong performance and high accuracy in correct detection of smishing messages. Models can identify suspicious messages due to the common format that smishing messages have. However, these types of technical solutions are not effective in an environment where attackers are constantly evolving.

At the message level, models like the S-Detector model[1] perform well because they identify recurring words and structural patterns in the suspected smishing message. More specifically this model uses a naive bayes classifier to analyze noun frequency patterns within incoming messages. By comparing old previously identified smishing messages to new incoming

messages, the model has shown promise. This machine learning solution has 96.4% detection accuracy in detecting new messages. More recent systems incorporate advanced natural language processing models and multi-layer detection architectures. For example, the Cybersmish framework integrates message analysis, URL inspection, and HTML content analysis to detect malicious campaigns with high accuracy[2]. Moving beyond just messages to HTML and URL content analysis play an important role since it better reflects the layered nature of modern smishing messages.

Smishing attackers frequently modify message templates, rotate and create new malicious domains, and use URL shortening services to evade detection systems. Due to frequency of changes, machine learning models require constant retraining and large datasets in order to be effective in practice. These limitations suggest that purely technical solutions, even if they are highly effective in detection, may not be sufficient to fully address the evolving nature of smishing threats. The problem is not whether models can classify the messages correctly, it's whether a machine learning team could keep its pace with adversaries who intentionally change tactics often. It's also important to note that multiple entities may conduct smishing campaigns so there will be a need to collect mass amounts of smishing data at all times to retest and retrain models.

Another important point is that technical effectiveness does not mean an automatic practical solution to the worldwide smishing issue. Some studies have noted language limitations, retraining demands, and the constant dependence of new datasets. Overall, a machine learning solution is necessary for the technical defense that needs to be incorporated but it's not a stand alone solution and has its own peculiarities. Its value is strongest when paired with other strategies that address the human aspect of smishing.

4. Effective defense requires effective education and easy reporting methods

Across many studies, cybersecurity education and awareness programs are consistently identified as a critical component of smishing mitigation strategies. This theme appears across both technical and user-centered studies which suggest that anti-smishing strategies fail when users assume they know what to do when they see a suspicious message. However, the literature shows that many users are still uncertain about how to verify messages, where to report them, and what aspects of a message can be determined trustworthy.

Njuguna et al. (2022)[4] concluded that no single technological method can mitigate 100% of the smishing threats that the public faces. Instead, effective mitigation strategies must combine both technological defenses with education and training incentives.

Earlier research on smishing by Yeboah-Boateng & Amanor (2014)[10] conducted a study when smishing was first an emerging topic, they found that there needs to be a strong emphasis on Security Education, Training, and Awareness (SETA) programs to help reduce the public risk

against smishing. Their findings also showed that many users failed to identify suspicious messages and lacked basic awareness of phishing indicators.

More recent research also highlights the need for improved reporting systems and user interfaces to help. Participants in several studies expressed frustration with the lack of clear reporting mechanisms for suspicious messages. Researchers have proposed integrating reporting features directly into messaging applications and increasing awareness of supporting services such as 7726, which allows users to report spam messages to mobile carriers.[8]

Together, these findings suggest that effective smishing defense requires a multi-layered approach that combines user education, improved reporting systems, and technical detection tools.

Discussion

This is the future directions and advice to others section - address limitations and talk about future projects that can build on this semester's work. You can describe some research gaps not addressed in the literature. That could be types of studies or types of tools that need to exist but aren't built, it could also be a type of research that is not commonly used but would be helpful to address the gap. Also it can be advice for security professionals and for government or military all of whom need insights for security awareness training and for procurement of end user tools. Finally you can point to specific studies in your review that show particular promise eg 'this group is on to something.

This literature review is meant to highlight a critical shift in how smishing should be conceptualized. Smishing is not a type of isolated or unorganized event, it is a type of phishing that is part of a coordinated cybercrime ecosystem. Threat actors integrate technical infrastructure, human manipulation, and an adaptive operational strategy to develop mass smishing attacks. Across this review, there are three key insights that form.

First, I notice that smishing campaigns demonstrate characteristics of organized and professional-like operations. Evidence from botnet behaviors, domain reuse, and consistent campaign coordination suggests that threat actors have a structured workflow and maintenance cycles. This often mirrors some form of a software engineering approach with a potential business model or idea. This alone challenges a traditional mindset where organizations and others treat or think about smishing as a low-level type of opportunistic fraud. While there are low-level, opportunistic, and rudimentary smishing attempts, that does not encompass the true value of the smishing problem. Smishing needs to be treated the same way alongside other forms of organized cybercrime, which will require many forms of defense including a technical and education aspect.

Secondly, I'd like to highlight that the literature reviewed constantly reinforces that human behavior is not a secondary or tertiary method to achieve smishing success. It is a primary indicator of identifying a smishing attack and a primary weapon used effectively when it

connects with susceptible people. Psychological triggers are purposefully used and are not incidental on a case by case basis. Smishing is a type of social engineering, and the psychology are deliberately engineered components in the smishing campaigns. This reframes the problem and changes the smishing issue from a purely technical to psychological in nature. Even if you have pure technical solutions, they are still not able to stop a user from interacting with messages when under emotional stress. A pure technical solution is an incomplete solution.

Lastly, there is lots of research that focuses on a newer machine learning approach that has lots of promise, but only in controlled environments. While the machine learning approach uses real smishing data, its real world effectiveness is constrained by the constant adversarial adaptation and reengineering of new content. The threat actors continuously change the structure of their messages, URLs, and their delivery of the message. Targeting smishing can be compared to shooting a moving target in many different environments. The machine learning models cannot keep up and continue to stay accurate without a constant stream of new data to train and refit models. A successful detection system needs to be able to target a constantly evolving threat. This means that models need to be chosen to have full-time support if we want to solve the technical aspect in smishing mitigation.

My analysis above only covers a fraction of the step towards a smishing solution. Several research gaps still remain. There is a clear lack of integration between technical detection systems and human-centered interventions (including education). The studies in these literature reviews tend to include only researchers that are in the computer science field, none from the psychology or education field. There is also limited work on cross-platform coordination where smishing studies that intersect with email phishing, voice phishing, and social media attacks are lacking. This fragments the smishing success because we are not getting the full picture of the scope of these attacks and not including other subject matter experts. It seems that researchers have defined the fact that smishing is a technical issue that relies on human psychology, but I have not seen research that includes psychologists or educational experts.

For practicality, these findings require a multi-domain and multidisciplinary strategy. For the cybersecurity experts, this includes combining machine learning detection and user interface design that communicates risk effectively. For psychologists, this includes studying human emotion and response with technological devices to understand how people react to receiving messages that bring out emotional responses. For educational experts, this means finding the best way to educate susceptible persons. For policymakers and government organizations, there is a need to standardize reporting mechanisms and invest in large scale anti-smishing campaigns. It's important to note that smishing is a low-cost type of attack that has a high-cost impact and should be considered a high-level information exploitation threat that targets people and organizations.

As a collective, future research should prioritize researching the ecosystem-level impact, interdisciplinary collaboration, and a long standing study that tracks how smishing evolves over time. There are already promising threat models and data (ex. SmishViz), working detection solutions, and designs that involve users improving the development of smishing defense.

Conclusion

Smishing has evolved into a coordinated, psychological driven, and technologically adept threat that operates across multiple domains. The literature has consistently shown that modern smishing campaigns are not isolated events or rudimentary. They are now resembling structured operations that utilize automation, brand and government agency impersonation, and cross-channel communication to increase their effectiveness. At the same time, human emotions are invoked such as urgency, trust, and authority remain key to the success of smishing attacks.

While machine learning models are effective at working as filtering and warning systems, the research highlights a major limitation. These systems struggle with keeping pace with the rapidly evolving tactics and messages used by the attackers. The constant changes in messages, websites, and URLs reduce the long term effectiveness of a purely technical solution.

As a result, effective defense against smishing demands a more integrated approach. The literature communicates the importance of user-centered designs, cybersecurity education, and improved reporting mechanisms. Educate and awareness programs, along with better device warnings and reporting tools can help reduce user susceptibility and improve overall resilience against the quickly evolving threat.

There are still research gaps that remain. There is a lack of interdisciplinary work and research that bridges technical cybersecurity solutions with behavioral and education strategies. Future research should focus on developing ecosystem level defenses that account for both technological and human factors. They should improve cross-channel monitoring to better identify coordinated attack campaigns.

Smishing isn't a small scale scam problem, nor is it a minor extension of phishing. It is a complex and evolving security challenge. Addressing it effectively and proportionally will require collaboration across multiple domains including technical experts, human-computer interaction, cyber-education, and public awareness initiatives.

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Priorities of work (notes for me)

1. Include more information from more sources

Abstract notes

Define the state of the world

Include facts or statistics to back the statement

Could include an example of a well known event

Include a however statement ... (problem with the state of the world)

What has to happen to solve the problem (hero statement)

Images

